



ASSESSMENT OF WILD SPINY LOBSTER STOCKS ON THE BRAZILIAN CONTINENTAL SHELF

BY

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ABSTRACT

The red spiny lobster *Panulirus argus* (Latreille, 1804) and the green spiny lobster *Panulirus laeviscauda* (Latreille, 1817) are the most abundant and commercially valuable lobster species on the Brazilian continental shelf. Though sold on the domestic market, the painted spiny lobster (*Panulirus echinatus* Smith, 1869) and the slipper lobster (Scyllaridae) are of little economic importance and have been the object of very little research. Brazilian spiny lobster fisheries have been in decline for many years due to growth overfishing, including the capture of undersized specimens (>50%). Furthermore, fishing berried females and older lobsters from deeper waters (≥ 50 m), compromising the reproductive potential of the species and increasing the risk of low recruitment and fishing collapse. The setting of quotas on catches is in itself insufficient to protect Brazilian lobster stocks. Management efforts should instead focus on the enforcement of control measures in order to increase lobster stocks. In this study we make several recommendations: (i) lobster traps should be furnished with escape gaps for undersized lobsters, (ii) the use of undersized lobsters (decoys) in traps should be avoided, (iii) gillnets (“caçoeira”) and artificial shelters (“marambaia”) should be eliminated, (iv) an optimal minimum legal size should be established (80 mm CL for all spiny lobster species and 85 mm CL for slipper lobsters), (v) a maximum legal size should be established (135 mm CL for *P. argus* and 100 mm CL for *P. laeviscauda* of both sexes), (vi) the capture of berried females should be prohibited, (vii) a closed season should be instituted to protect ovigerous females and ensure recruitment, (viii) establishment of marine reserves or a total closed season to protect the juveniles in nursery areas, (ix) a monitoring program collecting catch and effort data should be implemented to help assess lobster stocks adequately, and, (x) stratified random sampling must be implemented.

RESUMO

As lagostas *Panulirus argus* (Latreille, 1804) e *Panulirus laeviscauda* (Latreille, 1817) são as espécies mais abundantes e de maior valor comercial capturadas na plataforma continental do Brasil. Outras espécies capturadas, tais como, a lagosta pintada (*Panulirus echinatus* Smith, 1869) e a

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lagosta sapateira são de pouca importância econômica e têm sido pouco pesquisadas. A sobrepesca do crescimento devido às capturas de grandes números de ilegais (>50%) é a principal causa de rendimento decrescente em lagostas. Além disso, a pesca de fêmeas ovadas e lagostas grandes em águas profundas (≥ 50 m) poderiam deprimir o potencial reprodutivo da população, da espécie, e pode aumentar do risco de baixo recrutamento e colapso. A fixação de quotas de captura talvez não seja em si insuficiente para proteger o stock pesqueiro do Brasil. Os esforços de gestão devem concentrar-se na aplicação de medidas de controle, a fim de aumentar os estoques de lagostas. Neste estudo, se faz várias recomendações: (i) as armadilhas para lagostas devem ser equipadas com aberturas de escape para lagostas ilegais, (ii) se deve evitar o uso de lagostas por abaixo do tamanho mínimo legal (iscas) em armadilhas, (iii) devem ser removidas as rede de fundo (“caçoeira”) e refúgios artificiais (“marambaias”), (iv) um comprimento ótimo mínimo legal deve ser estabelecido (80 mm CL para as lagostas espinhosas e 85 mm CL para as sapateiras), (v) deve ser estabelecido o tamanho máximo legal (135 mm LC para *P. argus* e 100 mm LC para *P. laevicauda*) de ambos sexos, (vi) deve ser proibida a captura de fêmeas ovígeras, (vii) defeso total deve ser instituído para proteger as fêmeas ovígeras e assegurar o recrutamento, (viii) deve ser estabelecida uma área marinha protegida ou um defeso total para proteger os juvenis em os criadouros, (ix) devem ser estabelecida um programa de monitoramento de coleta de captura e de esforço para ajudar a avaliar estoques de lagosta adequadamente, e (x) deve ser estabelecida uma amostragem aleatória estratificada.

INTRODUCTION

Lobster fisheries on the Brazilian continental shelf target the red spiny lobster (*Panulirus argus* (Latreille, 1804)), the green spiny lobster (*Panulirus laevicauda* (Latreille, 1817)) and the painted spiny lobster (*Panulirus echinatus* Smith, 1869). Small numbers of slipper lobsters, including *Scyllarides brasiliensis* Rathbun, 1906, *Scyllarides delfosi* Holthuis, 1960 and *Parribacus antarcticus* (Lund, 1793), are caught incidentally during spiny lobster fisheries (Ivo et al., 1996). The southern slipper lobster *Scyllarides deceptor* Holthuis, 1963 is captured as bycatch in shrimp trawls off southern Brazil (Oliveira et al., 2008), while *Scyllarus chacei* Holthuis, 1960 is exceedingly rare on the Brazilian coast, with only one report of a berried female captured during shrimp fisheries off the state of Amapá (Silva et al., 2012).

Brazilian artisanal lobster fishermen currently use a single-entrance rectangular baited trap referred to as “manzuá” or “covo”. In the late 1980s, a larger trap with two entrances, referred to as “cangalha”, went into operation in some locations on the coast of Ceará (Fonteles-Filho, 2000). Traps consist of a wooden frame covered with a wire or nylon mesh and are legal in use (IBAMA, 2008). However, illegal gillnets (referred to as “caçoeira”) are still used to catch spiny lobsters (Moura, 1963). In addition, since 2000 artificial shelters of varying design (referred to as “marambaias”) have been deployed clandestinely on fishing grounds off northeastern Brazil (Nascimento, 2006). No fishing biology information is available for artificial shelters in this region.

The bycatch of Brazilian lobster fisheries is very diverse regardless of season and depth and includes fish, non-target lobsters, crabs and especially mollusks (Ivo

et al., 1996). Nothing is discarded: most of the bycatch is used to bait the traps or is consumed onboard, while the most valuable items, such as slipper lobsters, painted spiny lobsters and good table fish, are traded on the domestic market.

An overview of the past 500 years of Brazilian fisheries management was published by Paiva (2004), while Cavalcante et al. (2011) analysed the lobster fishery management plan currently adopted in Brazil. Spiny lobster fisheries are managed nationwide by IBAMA (Institute for the Environment and Natural Renewable Resources) through a set of measures. Thus, it is illegal to catch *P. argus* under 75 mm CL (130 mm tail length) and *P. laevicauda* under 65 mm CL (110 mm tail length), a closed season of six months (usually December-May) has been instituted, and nursery grounds up to three miles from the coast are permanently closed to lobster fishing. Fishing with gillnets and commercial scuba diving in natural and artificial shelters are forbidden, as is the taking of berried females (Fonteles-Filho, 2000). Fishing licenses have been limited since 2000 (Brasil, 2008).

The spiny lobster stocks on the Brazilian continental shelf have not been completely assessed yet and the impact of variations in the parameters determining fishing effort, catch and stock size is virtually unknown. The effects of errors in the current management strategy are not examined directly in this study, but an attempt was made to provide a general framework to investigate such effects. The study contributes to the current discussion in the field by assessing the situation of exploited spiny and slipper lobster stocks and by proposing effective control measures capable of protecting and recovering populations of target and non-target species. It is hoped that our findings and recommendations will stimulate further research with a view to establishing a sustainable lobster fishing management plan.

MATERIAL AND METHODS

Site description

Palinurids and scyllarids (with the exception of *Scyllarides deceptor*) are distributed throughout the Brazilian continental shelf from Amapá (04°26'N 51°32'W) to Espírito Santo (21°17'S 40°56'W) (fig. 1). Fishing grounds are numerous in the north (off Amapá and Pará), the northeast (off Maranhão, Piauí, Ceará, Rio Grande do Norte, Paraíba, Pernambuco, Alagoas, Sergipe and Bahia) and the southeast (Espírito Santo), covering a total surface area of 74 607 km² (Paiva, 1997) or 84 534 km² (Fonteles-Filho, 2000), depending on the source. Lobster habitats are diverse, including reef structures and calcareous algae beds (Coutinho & Morais, 1970) at a depth of 1-50 m, although fisheries of *Panulirus argus* at 50 to 100 m is reported to be developing in the north (Silva et al., 2003).

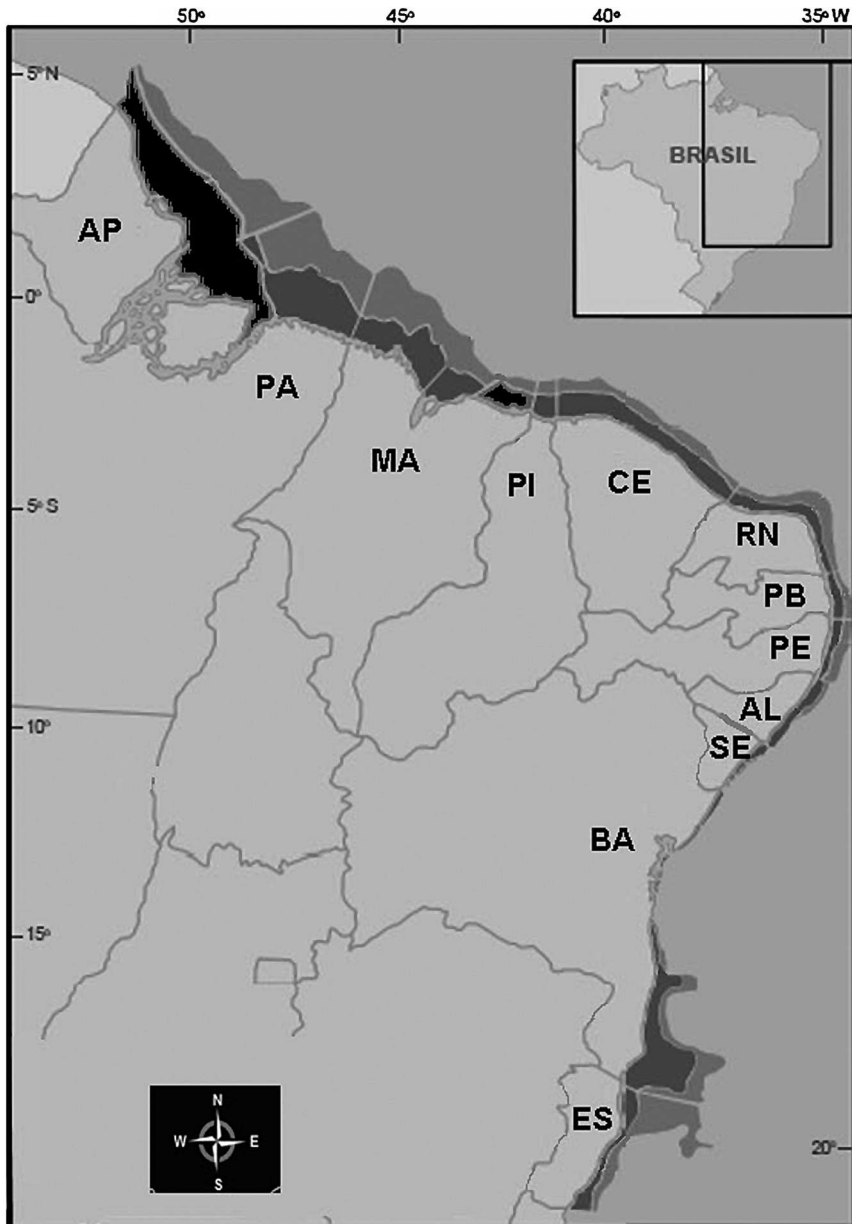


Fig. 1. Map showing the fishing area corresponding to each state along the Brazilian continental shelf. Amapá (AP), Pará (PA), Maranhão (MA), Piauí (PI), Ceará (CE), Rio Grande do Norte (RN), Paraíba (PB), Pernambuco (PE), Alagoas (AL), Sergipe (SE), Bahia (BA) and Espírito Santo (ES). Black = muddy bottom; dark grey = shallow waters (<50 m); light grey = deeper waters (≥ 50 m). Modified from Cruz et al. (2011).

On the other hand, *S. deceptor* is most widely distributed along the coast of southern Brazil, from Rio de Janeiro (21°17'S 40°57'W) to Rio Grande do Sul (33°44'S 53°22'W). The species is captured by the shrimp fleet as bycatch at a depth of 14-420 m (Oliveira, 2008).

Databases and field surveys

Brazilian field surveys of lobster traps were conducted for 113 fishing grids (30 × 30 m), most of which are located on the inner shelf (Fonteles-Filho & Guimarães, 1999). The depth distribution of *Panulirus argus* (1-100 m) and *P. laeviscauda* (1-50 m) was determined and estimates of relative abundance (catch per unit of effort) were obtained from the catch data and fishing effort (number of traps checked per grid). Spatial and temporal variations of both species were registered for the period 1974-1991. The effort was expressed as the number of traps checked per grid ($n \text{ km}^{-2}$) while the yield was expressed as the volume of lobsters landed per grid (kg km^{-2}).

Brazilian government agencies kept yearly updated databases of total lobster landings from 1965 to 2009. A time series of landings per species (*P. argus* and *P. laeviscauda*) is available for the period 1965-1990 (Ivo & Pereira, 1996). To estimate the volume of *P. argus* landed between 1991 and 2009 an empirical model was used with the linear equation:

$$y = a + b * x$$

in which $y = P. argus$ landings and $x =$ total lobster landings. The volume of *P. laeviscauda* landed (1991-2009) was calculated with the equation:

$$P. laeviscauda \text{ landings} = \text{total lobster landings} - P. argus \text{ landings.}$$

Moving average analysis was used to examine the trends in all the data sets (second period).

Information on slipper lobster (Scyllaridae) landings in northeastern Brazil (1999-2006) was obtained from CEPNOR (Center for Research and Management of Fishing Resources in Northern Brazil) and from IBAMA. The time series of *S. deceptor* landings were obtained from IBAMA (1991-1999) and from commercial fisheries bulletins published in Santa Catarina (2000-2010).

Stock productivity and fishing intensity

By definition, the stock is the set of exploited animals (Laurec & Le Guen, 1981). The parameter fish biomass (production) per unit of area and per unit of time was described by Ricker (1975), Minns (1997) and Randall & Minns (2002). It is usually calculated as the ratio between the product of commercial fisheries and

the area of fishing ground (Munro & Thompson, 1983). In this study, this ratio is referred to as the stock productivity index (S). It is given by the equation $S = C/A$ (Seber, 1982), where C is the weight (kg) of the captured animals and A is the area (km^2).

Fishing intensity (f_{int}) is defined as fishing effort per unit area per unit time. Hence, it is equal to the effective area covered by the gear per unit time as a fraction of the area occupied by the stock (Sanders & Morgan, 1976; Munro & Thompson, 1983). The equation used was:

$$f_{\text{int}} = \frac{\sum_{i=1}^n f_i}{A}$$

where the numerator is the sum of all f_i values from f_1 through f_n (n is the number of grids), divided by the area (km^2). In this study, f_{int} is the number of traps checked per year divided by the area of trap fishing ground (ATFG) per unit of time (years).

Stock productivity (S) and relative abundance (CPUE) were compared as measures of habitat productive capacity for lobster stock, based on CPUE data (catch per unit of effort) for spiny lobster trap fishing along the Brazilian coastline over a period of 18 years (1974-1991).

Furthermore, stock productivity (S) was calculated for each lobster species (*P. argus*, *P. laevicauda* and *Scyllarides* spp.), as proposed by Cruz et al. (2011), by dividing annual landings by the fishing area (km^2) of each state, using the equation: $(\text{Landings}_{\text{year}})/A$. Thus, annual S values were obtained for *P. argus* (isobaths < 100 m) for the states of Pará (PA) covering the years 1998, 1999, 2000, 2004 and 2005, Maranhão (MA) in 2003, 2005 and 2006, Ceará (CE) in 1999-2006, Rio Grande do Norte (RN) in 1999-2006, Pernambuco (PE) in 1999-2006, Alagoas (AL) in 1999, 2000, 2001, 2005 and 2006, and Bahia (BA) for the period 2002, 2005 and 2006. Likewise, S values were obtained for *P. laevicauda* (isobaths < 50 m) for PE (1999-2006), AL (1999-2006) and BA (2002, 2005, 2006), and for slipper lobsters (isobaths < 100 m) for MA (2005), RN (1999-2006) and PE (1999-2006) (fig. 1).

The plotted time series cover 46 years (*P. argus*), 16 years (*P. laevicauda*) and 17 years (*Scyllarides* spp.). To examine the tendencies between the S values, we found the best-fitting polynomial regression curve, where x is the time series of each state and y is the annual S value.

To calculate the S value for *P. argus* in the state of Pará, we used catch data from surveys conducted by the shrimp fleet along the north coast of Brazil (Silva et al., 2008) during the years 2001, 2002 and 2003. A total of 4900 (2001), 2128 (2002) and 1324 (2003) specimens were captured in this area ($12\,568\text{ km}^2$) at 80-104 m. The average carapace length (mm CL) was calculated and used to determine the

average weight, using the equation (Cruz, 2002):

$$W_t = 0.002582 \times CL^{2.7461}.$$

The total catch (kg) was estimated by multiplying the number of lobsters by the average weight. The annual S value was calculated as the mean catch (kg) per area (km^2) and year.

Surplus production models

Surplus production models were applied to data sets of different years and cells of the Brazilian continental shelf, at different levels of effort. The Fox (1970) model was applied to determine the level of effort per year per km^2 (f_{int}) which produced the maximum sustainable yield per area (MSYA) without affecting long-term stock abundance. Based on this model, the following equation was employed:

$$\ln \text{CPUE} = a + b * f_{\text{int}}$$

where CPUE = catch per unit of effort per fishing area, a = intercept, b = slope and f_{int} = the number of fishing devices checked by area. Thus, $f_{\text{op}} = -(1/b)$ (optimal fishing intensity), and $\text{MSYA} = -(1/b) \times \exp(a - 1)$.

The maximum sustainable yield (MSY) was obtained by multiplying MSYA by the area (km^2) of trap fishing ground (ATFG) per unit of time (years). Thus, $\text{MSY} = \text{MSYA} \times \text{ATFG}$.

RESULTS

Landings according to species and fishing gear

Since *Panulirus argus* represents 75% of lobster landings on the Brazilian continental shelf in the period 1965-1990, a significant linear relationship can be established between *P. argus* landings and total lobster landings for the same year ($n = 26$; $r^2 = 0.9020$, $P < 0.001$), using the equation: Red spiny lobster landings_(t) = $225.17 + 0.6752 * \text{total lobster landings}_{(t)}$. Subsequently, the volume of green spiny lobster landed was found by subtracting red spiny lobster landings from the total lobster landings (fig. 2).

The earliest Brazilian records of *P. argus* catches date from 1965 (2435 t). In 1974, catches had increased to nearly 7800 t, but in 1976 only around 3500 t was recorded. In the period 1976-1990, catches fluctuated between 3600 t and 8100 t. Beginning in 1986, landings rose gradually to a record peak of 8841 t in 1991. Between 1992 and 1998, catches declined to 4325 t, then apparently stabilized between 2001 and 2009, averaging 5026 t (fig. 2).

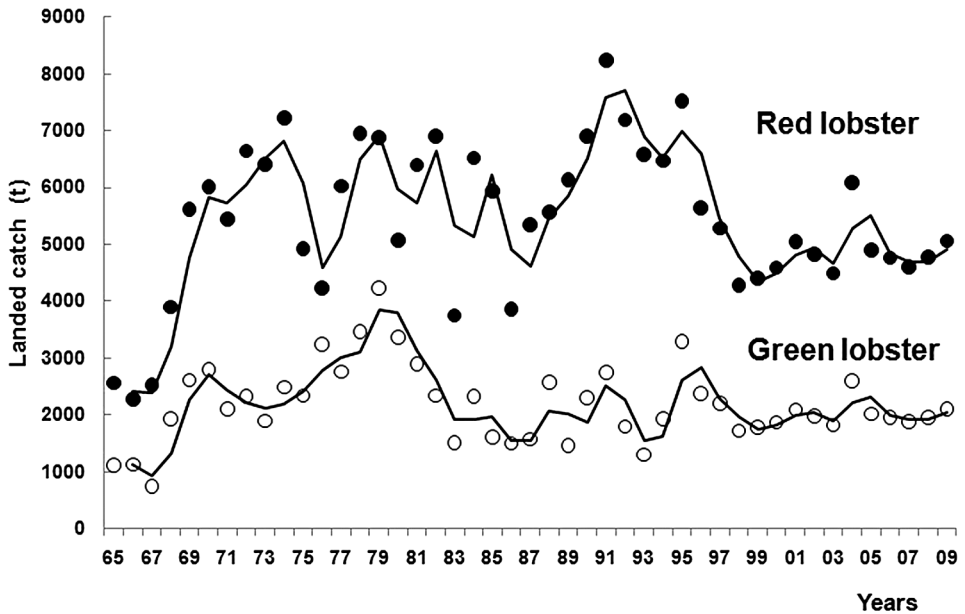


Fig. 2. Landing trends for red spiny lobster (*Panulirus argus* (Latreille, 1804)) and green spiny lobster (*Panulirus laeviscauda* (Latreille, 1817)) on the Brazilian continental shelf. Period 1965-2009. The line represents the moving average (second order).

Likewise, the earliest Brazilian records of *Panulirus laeviscauda* catches date from 1965 (1072 t). Landings fluctuated greatly, with peaks in 1970 (2658 t), 1978 (3639 t), 1987 (2356 t) and 1995 (3111 t), and dramatic decreases in 1975 (1073 t), 1984 (714 t), 1993 (1309 t) and 1998 (1676 t) (fig. 2). No significant association was observed between the time series of *P. argus* and *P. laeviscauda* landings between 1965 and 1994 ($n = 30$; $r^2 = 0.0222$; NS), although the two species were captured with the same gear.

Slipper lobsters are an important group in the Atlantic, although landings have always been very limited. Thus, total annual slipper lobster landings along the coast of Brazil in the period 1999-2006 amounted to 60-127 t (average: 97.83 t). The annual landings of *Scyllarides deceptor* range between 1.0 t and 24.0 t, although peaks of 71.5 t have occurred (2000 and 2001) (fig. 3).

The study of lobster fisheries is complex because of the diversity in fishing gear and techniques. No official information is available specifying catch according to fishing gear and vessel, but preliminary data reveal considerable diversity from state to state: BA ranks highest with 35%, followed by CE and PE (15-17%), RN (10%), PA and PI (5%), and AL and AP (2.5%). Currently, some 22 types of unselective fishing gear are used by artisanal fishermen, including different models of baited lobster traps ("armadilha", "covo", "manzuá", "cangalha"), traditional baited fish traps and bottom gill nets ("caçoeira"), which are no longer legal. Free

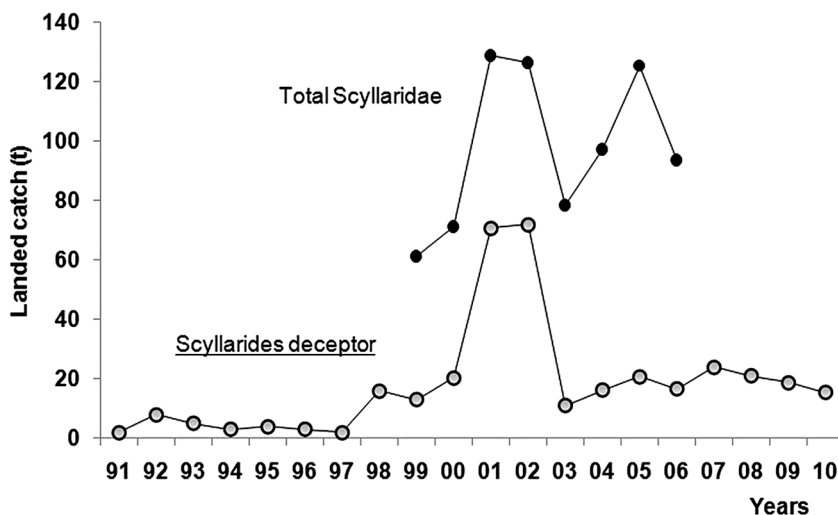


Fig. 3. Total landing trends for slipper lobsters on the Brazilian continental shelf (period 1999-2006). Landing trends for *Scyllarides deceptor* Holthuis, 1963 of southern Brazil (1991-2010).

divers and scuba divers in natural or artificial shelters catch lobsters by hand or with a hook. In addition, lobsters are occasionally caught with different types of purse seines, bottom trawls and line gear.

There is no official information on the number and fishing biology of the artificial shelters (“marambaia”) deployed off northeastern Brazil. Field observations suggest they are most common along the coast of CE ($2^{\circ}53'22''\text{S}$ $41^{\circ}13'46''\text{W}$ to $4^{\circ}06'06''\text{S}$ $38^{\circ}08'56''\text{W}$), RN ($4^{\circ}53'26''\text{S}$ $37^{\circ}14'37''\text{W}$ to $5^{\circ}12'06''\text{S}$ $35^{\circ}27'27''\text{W}$), PA ($6^{\circ}30'11''\text{S}$ $34^{\circ}57'30''\text{W}$ to $6^{\circ}57'57''\text{S}$ $34^{\circ}51'22''\text{W}$) and PE ($7^{\circ}33'07''\text{S}$ $34^{\circ}50'39''\text{W}$ to $8^{\circ}02'54''\text{S}$ $34^{\circ}52'42''\text{W}$). According to personal communications by local fishermen, artificial shelters may be built with a range of available materials, such as old car tires, multi-layered structures of flattened metal drums, scrap refrigerators, metal cans and wood or mangrove tree trunks.

When deployed, shelters of varying size are tied together in asymmetric groups and fastened to the bottom in areas with algal beds, sand or rocky surfaces at a depth of 3-60 m. To harvest the lobsters, the fishermen dive down to the shelter, with or without oxygen (supplied by a compressor connected to the boat’s motor). Each skin diver works from 0.5 to 1.5 h, with a total duration of 4 h. The lobsters are caught with a hook or by hand and kept in a bag. Some groups of fishermen collect lobster catch data which however are not made available to the public. According to many harvesters, artificial shelters yield more lobsters than traps.

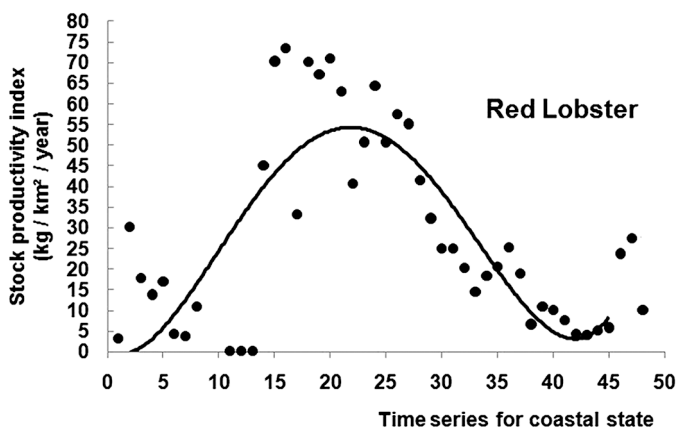


Fig. 4. Trends in the stock productivity index of *Panulirus argus* (Latreille, 1804) by region. Time series (1998-2005) from the states of Pará, Amapá, Maranhão, Ceará, Rio Grande do Norte, Pernambuco, Alagoas and Bahía. Polynomial analysis was used to obtain the best fit for the data (fourth power; $P < 0.001$; $r^2 = 0.6149$).

Stock productivity variability

The stock productivity index for *P. argus* varied greatly from state to state (fig. 4), with significantly higher values for CE and RN ($33\text{--}74 \text{ kg km}^{-2}$) than for any other Brazilian state ($<30 \text{ kg km}^{-2}$).

As for *P. laeviscauda*, the maximum S value was between 12 and 15 kg km^{-2} (fig. 5). S was greater for PB (average 8.71 kg km^{-2}) than for AL (average 2.51 kg km^{-2}) or BA (average 0.49 kg km^{-2}). Stock productivity for the slipper lobster

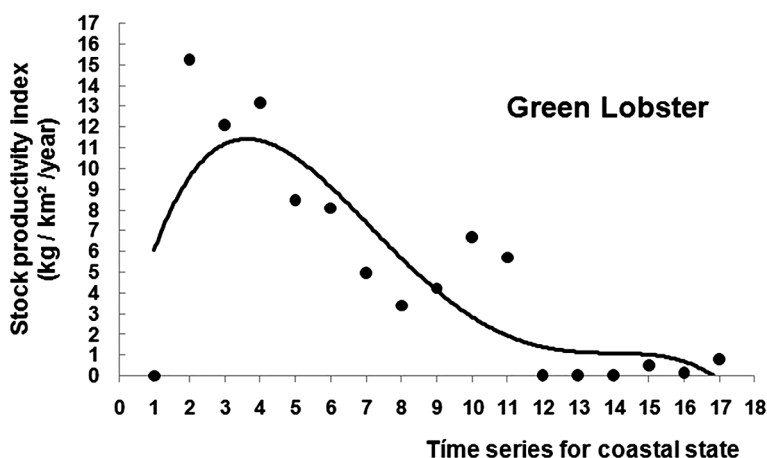


Fig. 5. Trends in the stock productivity index of *Panulirus laeviscauda* (Latreille, 1817) by region. Time series (1998-2005) from the states of Pernambuco, Alagoas and Bahia. Fourth-order polynomial curves were used to obtain the best fit for the data ($P < 0.001$; $r^2 = 0.6973$).

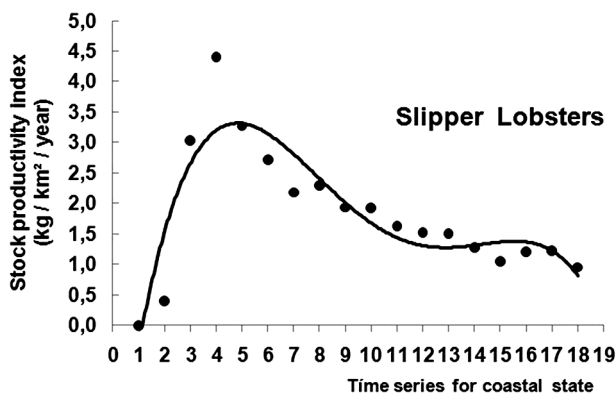


Fig. 6. Trends in the stock productivity index of slipper lobsters by region. Time series (1998-2005) from the states of Maranhão, Rio Grande do Norte and Pernambuco. Fourth-order polynomial curves were used to obtain the best fit for the data ($P < 0.001$; $r^2 = 0.7963$).

reached a maximum of 4.4 kg km^{-2} (fig. 6). S was greater for RN (average 2.72 kg km^{-2}) than for PE (average 1.30 kg km^{-2}). In the deeper waters (80-104 m) off PA, the S value for *P. argus* fluctuated from 0 to 0.31 kg km^{-2} , with an optimal maximum value of 0.25 kg km^{-2} (fig. 7), although the area features mostly older lobsters (average CL: $105 \pm 1.15 \text{ mm}$; 95% confidence limits) and over 90% of the females are sexually mature.

P. argus and *P. laeviscauda* differed significantly with regard to S ($P < 0.001$). In addition, within each species, highly significant differences were observed between

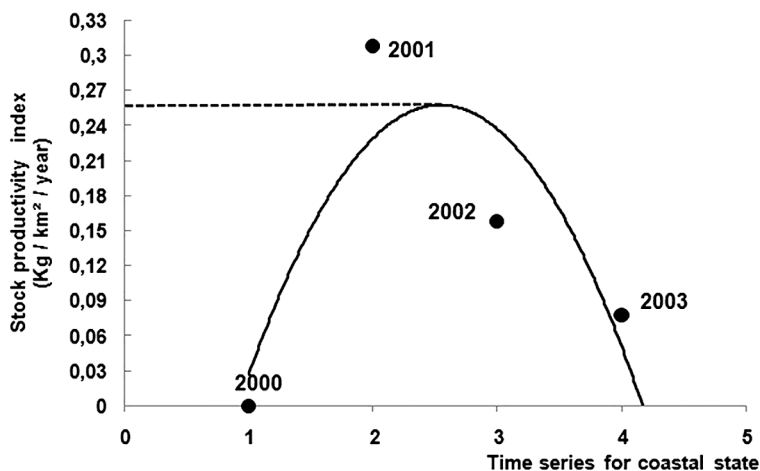


Fig. 7. Trends in the stock productivity index of *Panulirus argus* (Latreille, 1804) in regions with deeper waters (80-104 m), based on the number of red spiny lobsters caught during surveys conducted by the shrimp fleet between 2001 and 2003 off northern Brazil. Second-order polynomial curves were used to obtain the best fit for the data ($r^2 = 0.7321$).

periods ($P < 0.001$). Thus, the S value for *P. argus* ranged from 108 ± 23.48 (average $\pm$ 95% confidence limits) in 1974-1979 to 69 ± 10.88 in 1980-1989. The S value for *P. laeviscauda* ranged from 52 ± 16.58 (average $\pm$ 95% confidence limits) in 1974-1979 to 17.41 ± 3.29 in 1980-1989.

Model of lobster stock assessment

An equilibrium relationship between annual fishing intensity (f_{int}) and CPUE was determined by fitting regression curves for the period 1974-1991. The relationship was highly significant for *P. argus* ($n = 18$; $r^2 = 0.4962$; $P < 0.001$) and *P. laeviscauda* ($n = 18$; $r^2 = 0.7550$; $P < 0.001$) caught with traps on the Brazilian continental shelf. The following equations were used:

$$\ln \text{CPUE} = -0.0012 f_{\text{int}} - 2.4636 \text{ (} P. \text{ argus caught with traps)}$$

$$\ln \text{CPUE} = -0.0032 f_{\text{int}} - 2.5281 \text{ (} P. \text{ laeviscauda caught with traps)}$$

The estimate of MSYA presented here is satisfactory; the model was adjusted for rapid changes in f_{int} and marked fluctuations in year-class strength. The maximum sustainable yield per area was approximately 26 kg/km^2 for *P. argus*, attained with an optimal fishing intensity of approximately $833 f_{\text{year km}^{-2}}$ (fig. 8) in an average fishing area of $360\,819 \text{ km}^2$ (1974-1991). Likewise, a maximum sustainable yield of 9381 t may be assumed for the period 1974-1991. The S values plotted for 2008, 2009 and 2010 exceeded MSYA, with a f_{int} ($2239 f/\text{km}^2$) higher

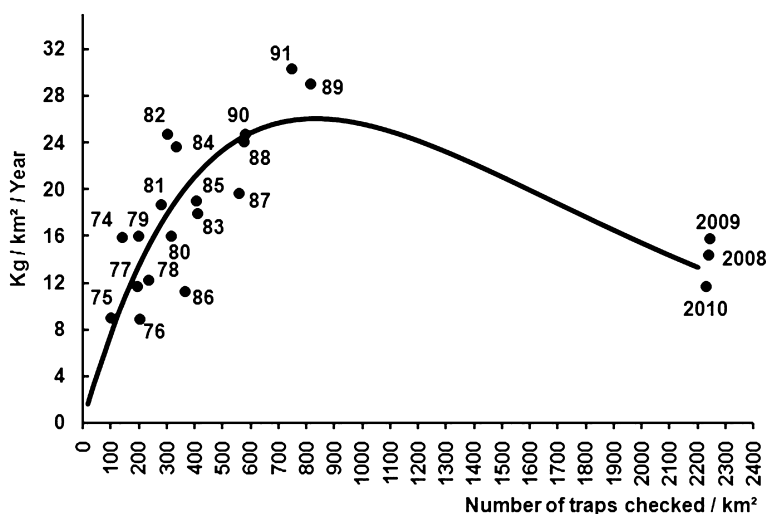


Fig. 8. Surplus yield curves for *Panulirus argus* (Latreille, 1804) based on annual trap productivity (kg km^{-2}) in relation to fishing intensity ($f_{\text{year km}^{-2}}$) on the Brazilian continental shelf (period 1974-1991).

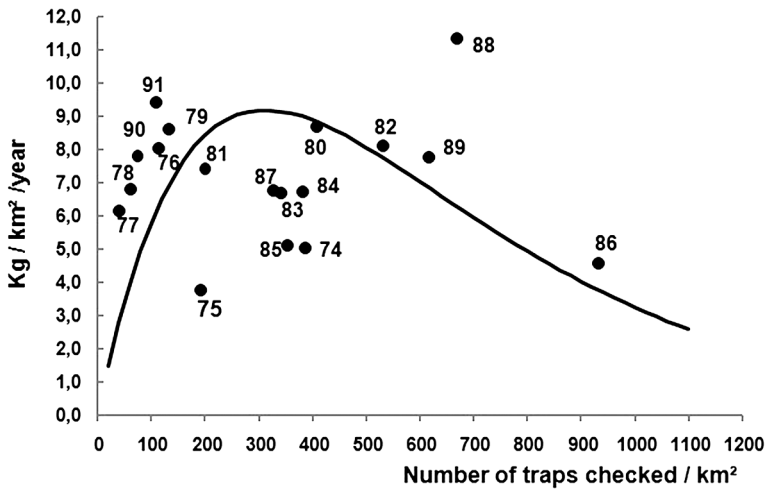


Fig. 9. Surplus yield curves for *Panulirus laeviscauda* (Latreille, 1817) based on annual trap productivity (kg km^{-2}) in relation to fishing intensity ($f_{\text{year}} \text{ km}^{-2}$) on the Brazilian continental shelf (period 1974-1991).

than the optimal fishing intensity. For the year 2010, the average S was estimated to be 12 kg/km^2 , with a MSY of 4330 t (12 kg km^{-2} for an area of $360\,819 \text{ km}^2$).

As for *P. laeviscauda*, the relationship between S and f_{int} (fig. 9) indicates a maximum sustainable yield per area of around 9 kg/km^2 attained by an optimal fishing intensity of approximately $313 f_{\text{year}} \text{ km}^{-2}$, with a MSY of 1473 t (9 kg km^{-2} for an area of $160\,510 \text{ km}^2$).

DISCUSSION

Trends in lobster stocks

Several local lobster populations, some commercially important (*Panulirus argus* and *Panulirus laeviscauda*) and some less important (*Panulirus echinatus* and slipper lobsters), are found on the Brazilian continental shelf. Due to closed geographic access, these lobster resources are exploited by the Brazilian fishing fleet only.

Recent studies have provided evidence that the populations of *P. argus* in the Caribbean Sea and on the Brazilian continental shelf are genetically and ecologically different (Diniz et al., 2005). In other words, Brazilian populations do not depend of foreign migrations of larvae. The same is most likely true for Brazilian populations of *P. laeviscauda*, *P. echinatus* and slipper lobsters. Cruz et al. (2011) hypothesized that current retroflexion in the western Atlantic (Patti et al., 2000; Krelling, 2008) can transport larvae hundreds of kilometers from spawning

areas into deeper areas settled by older lobsters, thereby mixing larval populations. Due to the extended duration of the phyllosoma stage in panulirids (Phillips et al., 2006), larvae may return as pueruli to the Brazilian coast carried by the current, thereby ensuring successful annual self-recruitment. If this is so, then the Brazilian populations of *P. argus* and *P. laeviscauda* are closed metapopulations, with no exchange with other Atlantic spiny lobster populations.

P. laeviscauda is common throughout the western Atlantic, but only populations on the Brazilian continental shelf sustain significant commercial exploitation. It is caught along with *P. argus*, especially in northeastern Brazil. Landings of *P. laeviscauda* in Pernambuco, where it is most abundant, account for 76 and 55% of national and world landings, respectively (Cruz et al., 2011). In this area, calcareous algae cover most of the bottom at 20 m isobaths (Coutinho & Morais, 1970), but so far little is known about the interactions between habitat and *P. laeviscauda* abundance.

Every year a large number of undersized spiny lobsters (>50%) are caught in shallow waters along the Brazilian coast as a consequence of excessive fishing effort (Fonteles-Filho, 2000) leading to severe growth overfishing and, consequently, declining yields, since the lobsters are harvested before they can reach a reasonable size. It would seem that growth overfishing and the interplay between environmental conditions (rain and ENSO) and pre-recruit abundance, as described by Cruz et al. (2011) but not yet fully explained, are the main causes of the large fluctuations in Brazilian spiny lobster landings.

A 5% decrease in the number of undersized lobsters taken by fishermen would increase abundance, the length at first capture, the yield per recruit and overall catches, as observed in Cuban lobster fisheries (Cruz et al., 1991). This fishing scenario is probably one of the determining factors the decline in Brazilian lobster landings.

Currently, there is no evidence of recruitment overfishing of *P. argus* and *P. laeviscauda* populations as a result of population reduction. However, the intensive exploitation of *P. argus* populations in deeper waters (50-100 m) since the 1980s (Paiva, 1997), targeting mainly older individuals (Silva et al., 2008), poses a serious risk to the reproductive potential and recruitment of Brazilian lobster stocks and could lead to a fishing collapse (Cruz & Bertelsen, 2009).

Despite being the main bycatch of lobster and shrimp fisheries in the Atlantic (Cruz et al., 2011), Brazilian slipper lobsters have been given little attention by fishing biologists due to their small commercial importance compared to spiny lobsters. Since lobster fisheries on the Brazilian continental shelf involve multiple species, it is essential to understand how intra- and interspecies interactions influence the abundance and distribution of populations in benthic ecosystems.

According to Cury et al. (2001), the key to answering these questions lies in understanding the energy flow within the ecosystems.

The sampling of Brazilian commercial lobster catches over the last 20 years has been very inconsistent. Further studies are needed to investigate the importance of different developmental stages (puerulus/juvenile/pre-recruit) in lobster fishing biology, and the effects of environmental factors on puerulus settlement of both spiny and slipper lobsters. Studies on lobster population dynamics are required to clarify the biological cycles, interactions and trophic flows of sympatric species which are key predators in the benthic ecosystem.

Lobster catches in deeper waters

According to Paiva (1997), Brazilian lobster fishing grounds cover a total area of 74 607 km². Fonteles-Filho (2000) estimated the fishing area by subregion: Northern Brazil = 17 508 km², Northeastern Brazil = 28 436 km², Southeastern Brazil = 12 661 km² and Southern Brazil = 24 948 km² (total = 83 553 km²). However, using geographic information systems, Cruz et al. (2011) calculated the total lobster fishing area between Amapá and Espírito Santo to be 356 610 km², of which 160 510 km² lies within a depth range of 1-50 m, and 196 100 km² is within a depth range of 50-100 m.

Ivo & Pereira (1996) and Silva (2003, 2008) described lobster fisheries in waters up to 100 m. In addition, several researchers and fishermen (personal communication) have reported the capture of specimens of *P. argus* at up to 150 m. Likewise, Silva et al. (2003) observed *P. argus* fisheries with gillnets deployed at 50, 60, 70, 80 and 104 m off the coast of Northern Brazil. Despite the low *S* values in this region (maximum: 0.25 kg km⁻²), reproductive activity is known to increase from near-shore areas towards offshore reefs, as described for the Wider Caribbean by Evans et al. (1995), Sutcliffe (1952, 1953), Kanciruk & Herrnkind (1976), Gregory et al. (1982) and Cruz & Bertelsen (2009). These areas may constitute an “optimal environmental window” (Cury & Roy, 1989), with ideal conditions for reproduction, to which lobsters migrate periodically from shallower waters, thereby increasing oceanic larval recruitment (Cruz & Bertelsen, 2009).

In deep-water habitats, *P. argus* is ecologically dominant due to its large size and feeds on a wide scope of benthic and infaunal species (Silva, 2012). When lobsters are abundant, they strongly influence prey abundance and composition. In some habitats, immigrant lobsters may even be killed and consumed by resident lobsters (Barkai & McQuaid, 1988).

The spacial-temporal growth of the lobster fleet between 1974 and 1979 entailed an expansion in the area of effective fishing ground to 45-35°W and 0-6°S, covering an area of approximately 44 415 km². Since 1980, CPUE has been in

decline and fisheries have expanded to northern and southeastern Brazil (48-34°W and 2°N-17°S), covering an area of approximately 213 286 km² (Cruz et al., 2011). This expansion of the grid area (360 819 km²) represents a 61% increment of the area (83 553 km²) previously calculated by Fonteles-Filho (2000). This is clearly reflected in a fishing trend perpendicular to the coast at a maximum depth of 50-100 m and a reduction in productivity from the coastal zone towards the deeper regions (Fonteles-Filho, 1997).

Different species of spiny lobsters (e.g., *P. argus*, *P. laeviscauda* and *P. echinatus*) can coexist in the same habitat, but may differ with regard to depth distribution. Thus, on the Brazilian continental shelf, *P. laeviscauda* is found at up to 40 m (Sousa, 1987) and exploratory surveys have confirmed its absence at depths greater than 50 m (Silva et al., 2003). On the other hand, the relatively scarce *P. echinatus* is reported to occur only in shallow waters between 1 and 25 m and usually in association with reef structures (Melo, 1999).

State of exploited stocks

Surplus production models are rarely used due to the high number of assumptions required to validate them (e.g. $F = q \times f$, where q remains constant but is usually a function of time, and resources are for the most part distributed homogeneously). However, it is also possible to consider the model as merely empirical because of the poor quality of the size structure and fishing data. In addition, we applied the considerations of Munro & Thomson (1983) to the model, where catch and effort are calculated per unit of area per year, and it is assumed that all units can support a similar level of production.

The estimate of maximum sustainable yield per area (26 kg km⁻²) between 1974 and 1991 presented here must be regarded as the first estimate made based on Brazilian lobster traps. The previous estimates of MSY for *P. argus*, e.g., 6095 t (Fonteles-Filho, 1992) and 6993 t (Fonteles-Filho, 2007) for a fishing ground area of 360 819 km², are lower than the estimates of the present study (9381 t) for an area of 356 610 km², as calculated by Cruz et al. (2011). However, since Brazil has the most extensive lobster fishing grounds in the Atlantic, but inconsistent catch data and little law enforcement, then in theory is feasible to obtain a red lobster harvest production between 9000 and 10 000 t.

Interestingly, the high variation observed in the catch of *P. argus* (between 3755 t and 8248 t) is believed to reflect fluctuations in catches of undersized lobsters and recruitment to the catchable stock. Nevertheless, it appears that the MSY levels of the period 2000-2009 (approximately 5000 t or 14 kg km⁻²) is close to the average landings in the same period (4989 ± 293; 95% confident limits).

The MSY of *P. laeviscauda* has previously been estimated to be 2460 t (Fonteles-Filho, 1979), 2285 t (Fonteles-Filho, 1992) and 2744 t (Fonteles-Filho, 2007).

However, more recent MSY estimates (1743 t) are less optimistic, indicating a decline due to intensive exploitation and stock depletion.

Brazilian multispecies lobster fisheries produce varying amounts of *P. argus* and *P. laeviscauda* along the Brazilian continental shelf using the same gear and unit of effort. Thus, the estimated potential catch of spiny lobsters for the period analyzed (1974-1991) may be assumed to be the sum of the MSY for *P. argus* (9381 t) and *P. laeviscauda* (1473 t), i.e., 10 854 t. Currently (2000-2009), the figure would be approximately 6500 t. The introduction and enforcement of catch limitations is recommended to prevent an increase in Brazilian lobster fishing pressure until more knowledge of the dynamics between fishing effort and lobster stocks has been obtained.

The estimates of MSY presented here should be regarded as tentative. The data on catch and effort by depth, from which the estimates were derived, need to be reevaluated through random lobster sampling along the Brazilian continental shelf, as suggested by Cruz et al. (2011). Furthermore, while the yield is known to be greatly affected by recruitment, no real measures of recruitment are available for the Brazilian continental shelf. A cursory examination of commercial length frequencies for the period under consideration suggests that increasing average size is an indicator of decreasing levels of recruitment (Cruz et al., 2011). Previous experience in the Cuban archipelago shows that decreasing abundance of pre-recruit and juvenile lobsters tends to increase the catchability of older lobsters (>4 years) (Puga et al., 1995).

To improve Brazilian spiny lobster fisheries and prevent a fishing collapse, a number of changes may be recommended. Thus, the deployment of fishing gear to catch older lobsters in deeper waters (50-100 m) and the capture of undersized specimens and berried females, compromising egg production and recruitment, should be prohibited. The setting of quotas on catches is in itself insufficient to protect Brazilian lobster stocks. Management efforts should instead focus on the enforcement of control measures in order to increase lobster stocks. In addition, lobster traps should be furnished with escape gaps for undersized lobsters, slipper lobsters and other forms of by-catch.

Management implications

Brazilian lobster fisheries have been regulated in some regions for more than a half century, but control measures are inconsistent and law enforcement is often ineffective. Present regulations include a minimum legal size and a closed season (Cavalcante et al., 2011).

The purpose of this study was to provide information for the management of this valuable but overexploited resource and suggest the adoption of new regulatory

measures to increase the reproductive potential and ensure sustainable recruitment of all lobster species, in the hope of reverting the effects of long-term overfishing.

Thus, our recommendations, along with some of the recommendations made by Cruz & Bertelsen (2009) for the Wider Caribbean but likewise applicable to lobster fisheries in Brazil, include: (i) lobster traps should be furnished with escape gaps for undersized lobsters and other forms of bycatch, (ii) gillnets (“caçoeira”) and artificial shelters (“marambaia”) should no longer be deployed; the latter may increase the exploitable but not the total biomass by aggregating juveniles, thereby leading to recruitment overfishing in regions with weak enforcement (Polovina, 1991), (iii) the capture of older lobsters in deeper waters (>50 m) should be prohibited (Ivo & Neto, 1996) to protect the productive potential and avoid disturbing sensitive habitats, (iv) establishment a maximum legal size of 135 mm CL for *P. argus* of both sexes; females may carry more than 1.5 million eggs (Cruz & Bertelsen, 2009) and spawn at least twice at an interval of 3-4 weeks (MacDiarmid & Kittaka, 2000), (v) establishment a maximum legal size of 100 mm CL for *P. laeviscauda* and *P. echinatus* of both sexes in order to protect the population reproductive potential, (vi) the practice of using undersized lobsters as decoys in traps should be discontinued, (vii) a minimum legal measure of tail width should be defined, (viii) the capture of berried females should be prohibited, (ix) a closed season should be instituted to protect ovigerous females and ensure recruitment, (x) an optimal minimum size should be established: 80 mm CL for all spiny lobster species and 85 mm CL for slipper lobsters, including *Scyllarides deceptor* (cf. Oliveira et al., 2008), *S. brasiliensis*, *S. delfosi* and *Parribacus antarcticus*, (xi) establishment of marine reserves or a total closed season to protect the juvenile in nursery areas, (xii) a monitoring program collecting catch and effort data should be implemented to assess lobster stocks adequately and (xiii) stratified random sampling must be implemented.

Over the past two decades the sampling of Brazilian commercial lobster catches has been inadequate and it is now in need of urgent improvement. In order to make meaningful assessments, adequate commercial catch and effort data should be collected regularly and properly. Without information on the processes that determine recruitment and its relation to fisheries and the climate, we cannot hope to understand the processes regulating the Brazilian lobster stocks.

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